SIXT HACKATHON 2026 · PROJECT 2 · THE GOLDEN BUTTERFLY

SIXT values and defleets cars on mileage it often doesn't know yet

67 days

median gap between a car's last known reading and the date we must predict (1–147 days)

- Long-term rentals report up to weeks late
- Some reports are wrong: typos, unit mix-ups
- No labels exist for wrong reports

TASK A

Forecast

The km at any date, up to ~5 months ahead

TASK B

Detect

Flag erroneous mileage reports, without labels

APPROACH

One engine for both tasks: predict km per day, not the odometer

km(t) = last reading + predicted km/day × days since that reading

Forecast · LightGBM

- Learns km/day from what is known at the cutoff
- Trained by replaying the past: a pretend “today” every 30 days
- Tested on vehicles never seen in training

Detection · same expectations

- Physical rules
- Check against the readings before and after
- Real-time mode using the forecast

KEY INSIGHT: BUSY CARS REPORT MORE OFTEN

29 vs 83

median km/day, cars with ≤ 10 vs 31+ report days (Jul–Nov 2025)

Weighting each car by its true number of targets fixed a **−400 km bias** (−430 → −194 km)

No leakage: actual return dates not used · test data never used for training

FORECAST RESULTS · HELD-OUT VEHICLES

16% lower error than the best naive baseline

1,416 km

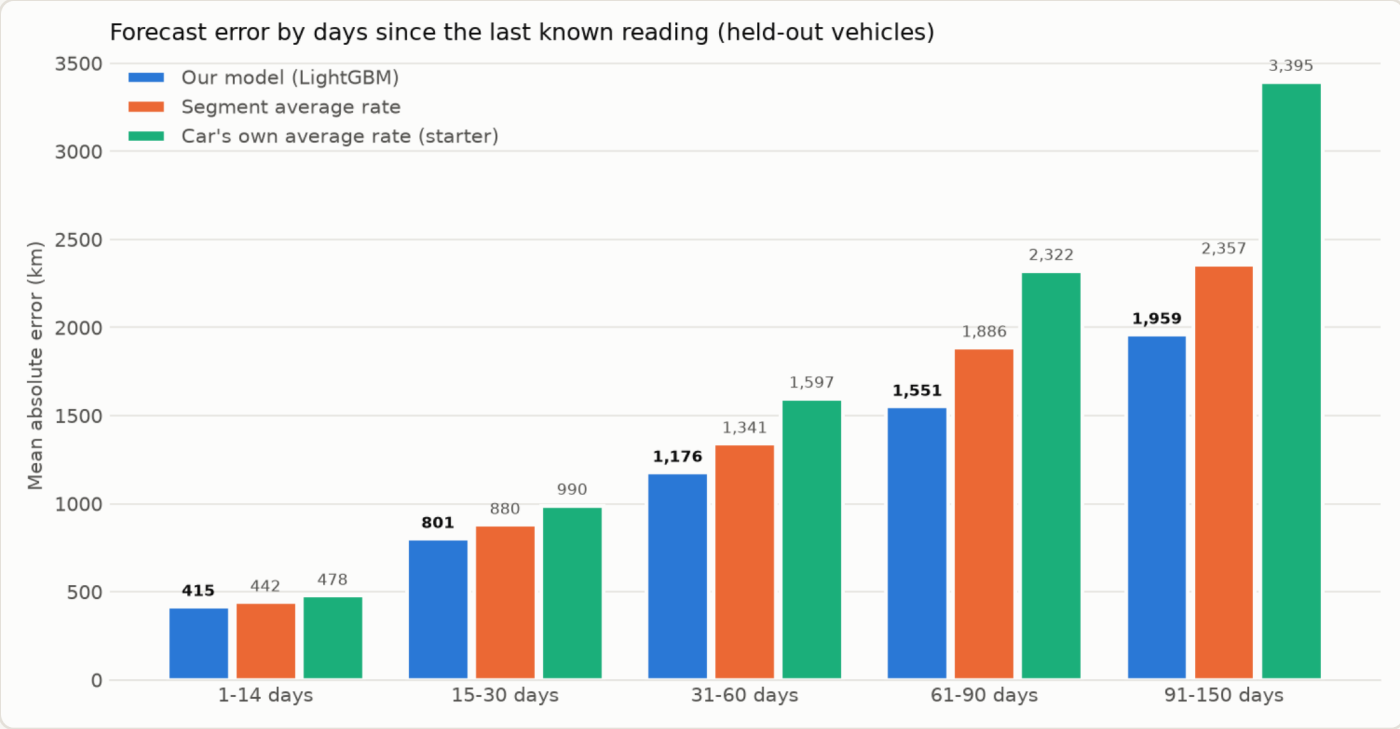
mean absolute error (MAE)

−16% vs segment average (1,680)
−36% vs starter baseline (2,200)

−48% when the last reading is older than 28 days (monthly reporting)

82% real coverage of our 80% uncertainty bands

3.6% error on the total km of the whole held-out fleet



The gain holds at all 3 validation periods (−12% to −16%) and grows with the horizon.

ERRONEOUS MILEAGE DETECTION · VS THE STARTER RULES, ON PLANTED ERRORS

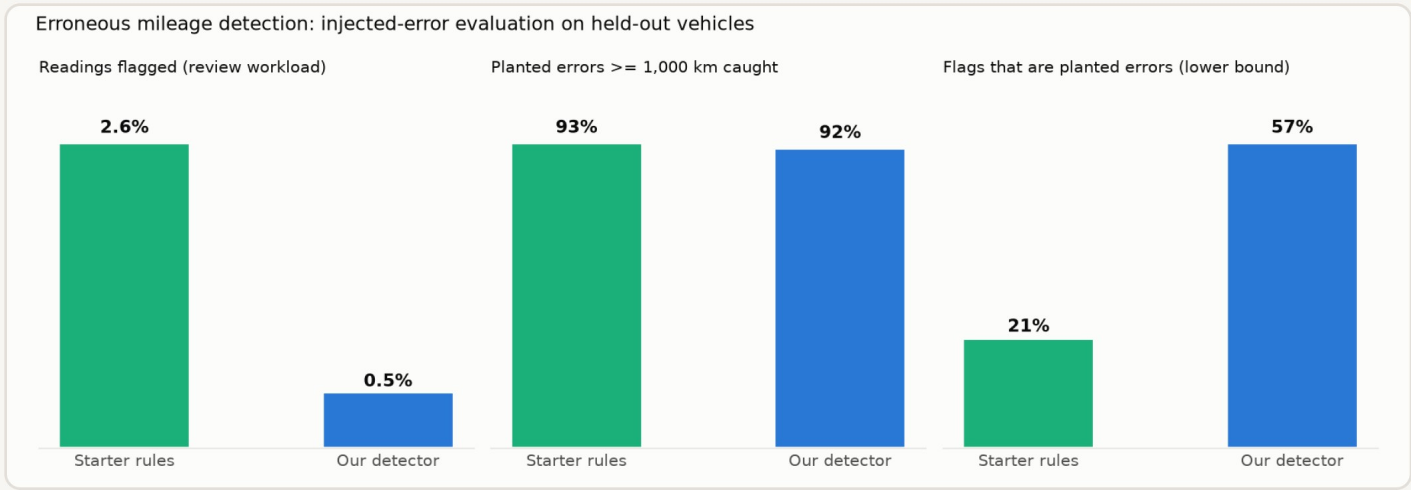
Same catch rate, 5x fewer alarms

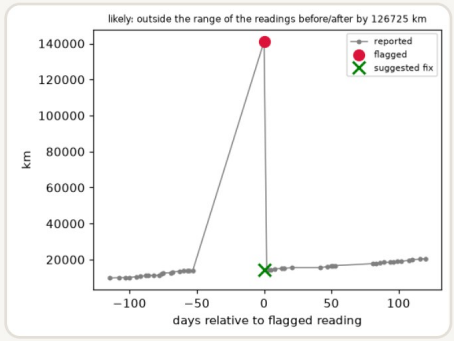
0.5%

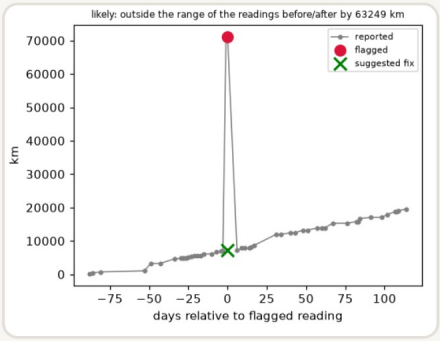
of readings flagged, vs 2.6% for the starter rules

92% of errors ≥ 1,000 km caught (starter rules: 93%)

97–98% right value suggested for swapped digits and miles-as-km







ORGANISERS' TEST DATA

1,018 flags

551,098 km between 55,074 and 55,412 → fix **55,109**

LIMITS AND NEXT STEPS

The missing signal is future rental activity, and SIXT already has it

Honest limits

- The remaining error is future rental activity: a linear model is only 5% behind, ensembles add 0.3%
- Stuck odometers can't be detected from mileage alone
- Test predictions are unscored

Next steps

- Plug in SIXT's booking calendar
- Review queue whose decisions become labels
- Request check-in readings from the most uncertain cars
- Score every incoming report in real time

ASK WHERE IT MATTERS

8.4%

of total error removed by a 60-day check-in from the 20% widest-band cars, vs 5.8% for 20% random cars